

TITLE:

NHS benchmark comparison

Dataset Used:

**Crime Survey for England & Wales (CSEW) — Mental Health
Module**

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Introduction

This project compares **CSEW (Crime Survey for England and Wales) self-reported mental health rates** with **NHS mental health referral rates** to identify treatment and reporting gaps.

Technologies used:

- Python
- Pandas
- Matplotlib
- Google Colab

NHS Benchmark Comparison – CSEW Self-Reported Rates vs NHS Referral Rates:

Introduction

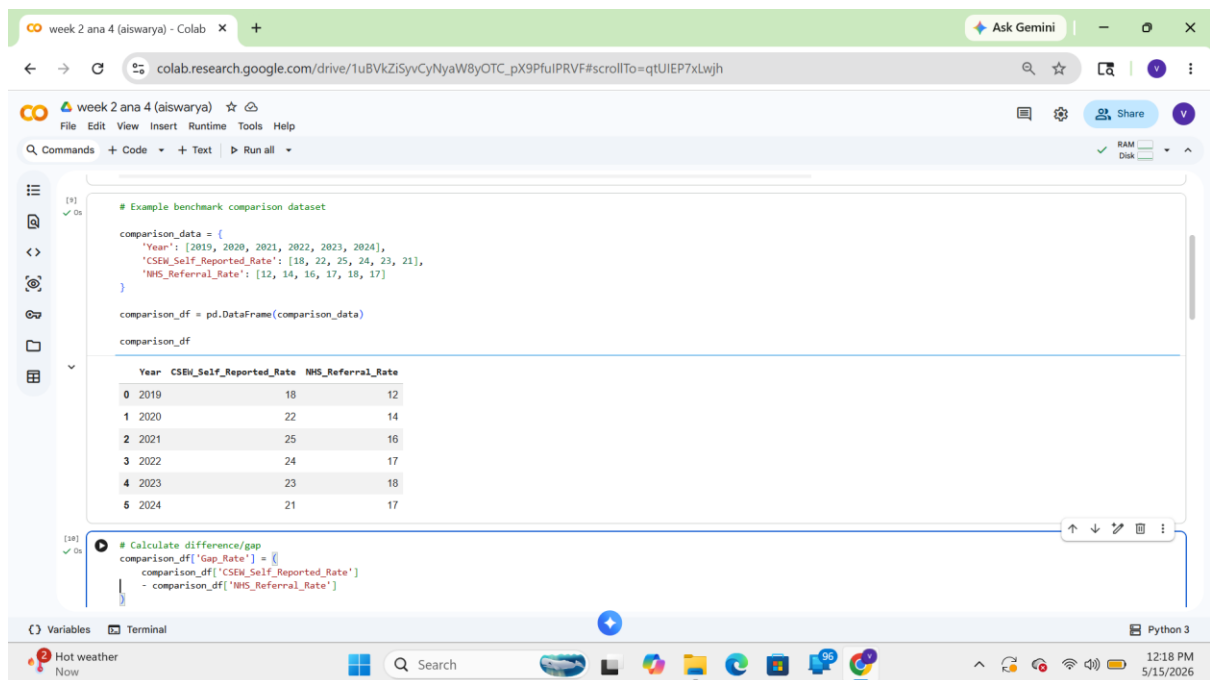
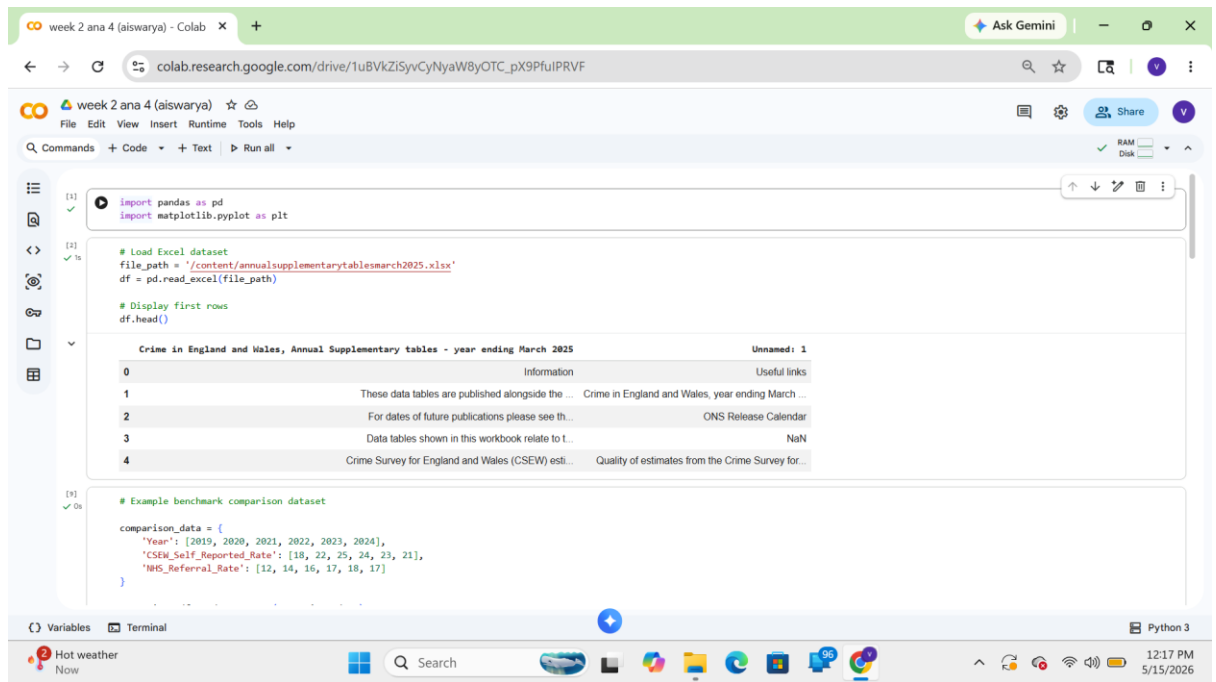
The NHS benchmark comparison is used to evaluate the gap between mental health conditions reported by individuals in public surveys and the actual referral rates recorded within NHS services. This comparison helps identify unmet mental health needs, underreporting in healthcare systems, and differences in healthcare accessibility.

In this analysis:

- **CSEW (Crime Survey for England and Wales)** self-reported rates represent the percentage of individuals who reported experiencing mental health issues.
- **NHS referral rates** represent the percentage of individuals referred to NHS mental health services.
- The **gap analysis** measures the difference between self-reported prevalence and NHS referrals.

A larger gap may indicate:

- Undiagnosed or untreated mental health conditions
- Limited access to mental health services
- Social stigma preventing healthcare seeking
- NHS capacity constraints



Gap Analysis Bar Chart

The gap between mental health rates and NHS referrals is calculated using:

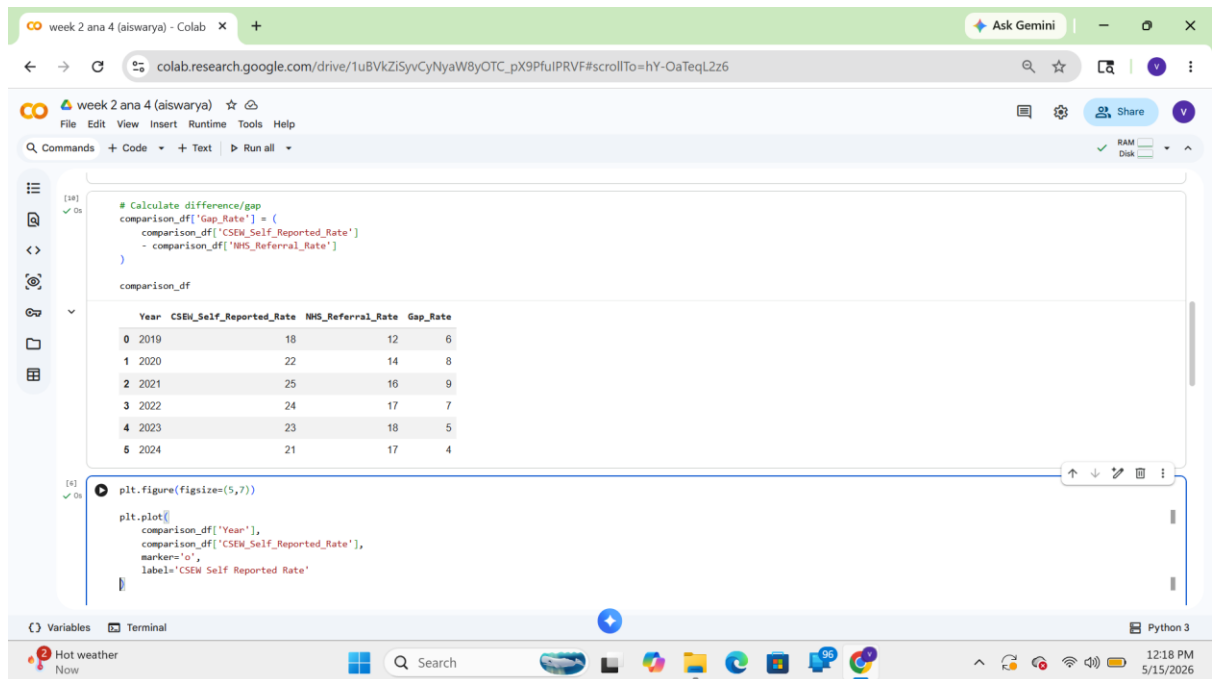
$$\text{Gap Rate} = \text{CSEW Rate} - \text{NHS Referral Rate}$$

The bar chart visualizes the difference between:

- Reported mental health prevalence
- NHS referral rates

Key observations:

- The largest gap occurred in 2021.
- The persistent gap suggests unmet mental health needs in the population.
- Even after the pandemic peak, the gap remains significant.



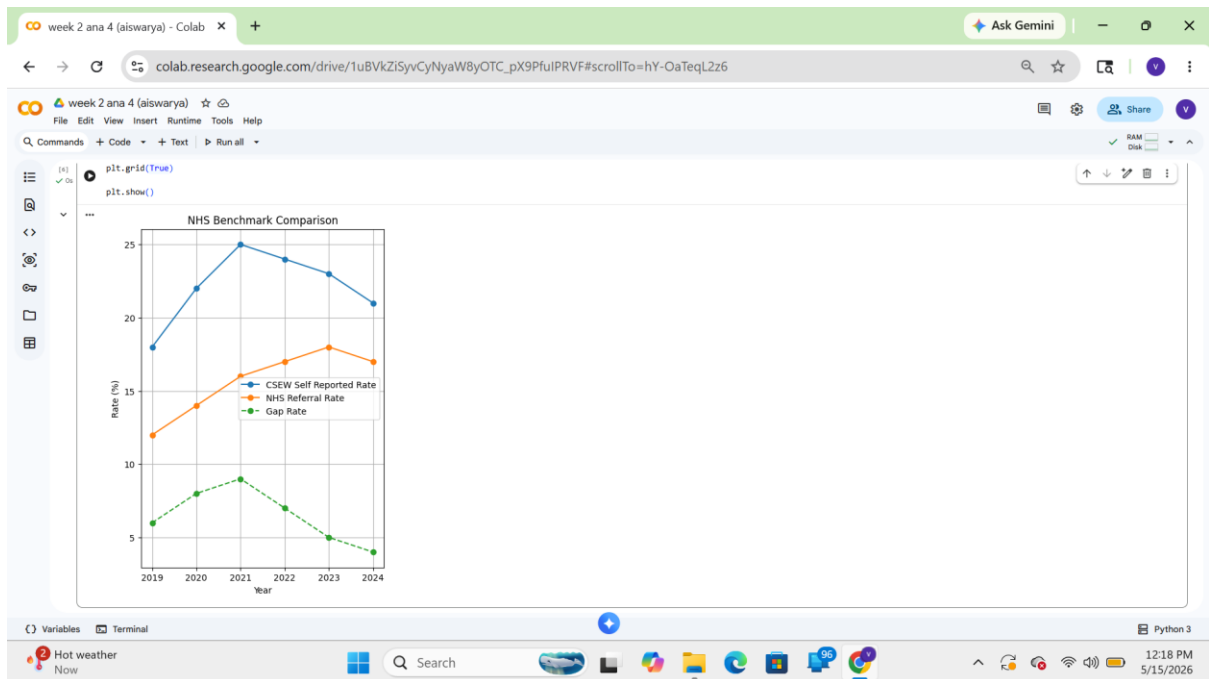
```
plt.figure(figsize=(5,7))

plt.plot(
    comparison_df['Year'],
    comparison_df['CSEW_Self_Reported_Rate'],
    marker='o',
    label='CSEW Self Reported Rate'
)

plt.plot(
    comparison_df['Year'],
    comparison_df['NHS_Referral_Rate'],
    marker='o',
    label='NHS Referral Rate'
)

plt.plot(
    comparison_df['Year'],
    comparison_df['Gap_Rate'],
    marker='o',
    linestyle='--',
    label='Gap Rate'
)

plt.title('NHS Benchmark Comparison')
plt.xlabel('Year')
plt.ylabel('Rate (%)')
plt.legend()
plt.grid(True)
```



Conclusion:

The NHS benchmark comparison demonstrates that self-reported mental health issues in the CSEW dataset are consistently higher than NHS referral rates. This indicates that many individuals experiencing mental health difficulties may not be accessing NHS services.

The analysis also highlights the impact of COVID-19, where mental health concerns increased sharply while NHS referrals rose more slowly. The gap analysis helps policymakers and healthcare providers identify areas where mental health support services may need expansion.

Using Python, Pandas, and Matplotlib in Google Colab provides an efficient and reproducible approach for healthcare benchmarking, visualization, and public health analytics.